TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Jung Han Choi	Email	jung-han.choi@hhi.fraunhofer.de
Institute Name	Fraunhofer Herinrich-Hertz Institute (HHI)			
Address	Einsteinufer 37, 10587 Berlin, Germany			

2. Technology Overview

Title of Proposed Technology	Optical front-end components (photonic-integrated circuit, photodiode) and their ASICs for Quantum-Key-Distribution (QKD) Communication			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum (√)			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Electrical · Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Information Security	Common Security Technologies	Optical Application Devices	Optical Components
Concept of Proposed Technology	The proposed technology proposes an optical front-end components (photonic-integrated circuit (PIC), photodiode) and their ASICs for Quantum-Key-Distribution (QKD) Communication. ● **Key Innovations** 1. High-sensitive SPAD and its ASIC: ✓ O- and C-band InP SPAD ✓ CMOS Active-Quenching ASIC ✓ Module construction 2. Mach-Zehnder based PIC and its co-designed driver ASIC ✓ InP-based Mach-Zehnder modulators for ALICE optical front-end ✓ Co-designed SiGe HBT driver ASIC			

Rationale for Joint R&D Project	● **Technical Necessity** - Recent advancement of Quantum QKD communication requires high-sensitive single-photon avalanche-detector (SPAD) and low-noise active quenching IC. The technological bottleneck lies in the SPAD fabrication and design technology of active quenching-IC. These Bob front-end components and their supply chain are of great significance. - Also, for Alice front-end components the reliable PIC fabrication and its low-power driver electronics are considered one of difficult technologies to achieve. The supply chain for these components in the world is not yet fully matured. - By having an internal collaboration with Fraunhofer, Korean industries can access modern QKD component technologies in EU. ● **Development Timeline and Cost Optimization** - Joint R&D with Fraunhofer Institute reduces development efforts (infrastructures for fabrication and characterization, etc.) considerably, approximately 40-50% by a whole. - According to survey, korean partner(s) alone cannot achieve QKD components. ● **International Collaboration Benefits** - Secured access to Fraunhofer's proprietary QKD photonics fabrication processes and characterization infrastructure - Enhanced global competitiveness and technology visibility through association with a world-class research institute - Accelerated commercialization pathway through European market channels and global certification frameworks
Key Words	1. Quantum QKD (Quantum Key Distribution) 2. Alice front-end components, InP PIC, ASIC 3. Bob front-end components, InP SPAD, ASIC

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	● **Primary Objective** Develop and demonstrate Quantum QKD Alice and Bob modules. ● **Specific Technical Objectives** 1. Alice and Bob PIC, their ASIC ✓ Design and fabricate InP photonic components ✓ Design and fabricate ASIC 2. Alice and Bob module ✓ Design and fabricate ALICE module ✓ Design and fabricate Bob module ✓ QKD module test			
R&D Target Specifications	1. Develop and design of InP PIC for Alice and InP SPAD for Bob 2. ASICs for Alice and Bob 3. Modules for QKD front-end 4. QKD module test			
R&D Plan	● **Phase 1: Feasibility Verification (6 Months)** ■ Objective: Demonstrate proof-of-concept for QKD front-end modules for Alice and Bob ■ Key Tasks: ✓ Define detailed component requirements and target specifications of each components ✓ Feasibility study and architecture simulation of ASICs for Alice and Bob ● **Phase 2: Prototype Development and packaging (30 Months)** ■ Objective: Develop and validate a TRL 5-level prototype of QKD front-end modules (Alice, Bob) ■ Key Tasks: ✓ Fabricate and characterize PIC, SPAD ✓ Fabricate and characterize active CMOS quenching-IC and driver ASIC ✓ Demonstrate and test using PAM-8 modulations.			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	■ Design, simulate, and fabricate PIC, SPAD ✓ PIC/SPAD design, simulation, fabrications ■ Design, simulatie, and fabricate ASICs ✓ Driver ASIC for Mach-Zehnder modulator ✓ CMOS Active quenching IC for SPAD
Expected R&R of Korean SME Partner	■ QKD front-end module design and fabrication ✓ Design, simulate, fabricate Alice module ✓ Design, simulate, fabricate Bob module ■ Prototype module test ✓ QKD Test platform ✓ Module test using QKD test platform

5. Expected Outcomes/Impact

Expected Outcomes and Impact	● **Technological Achievements** ■ Direct Outcomes ✓ World-best Quantum QKD front-end modules ✓ Early market introduction into world-wide QKD market ✓ Assuring supply chains ■ IP and Knowledge Outcomes ✓ High-impact patent applications ✓ Peer-reviewed publications ✓ Design documentation, technology transfer ● **Industrial and Economic Impact** ■ Technology Leadership ✓ Establishes Korea as global leader in high-reliable Quantum QKD ✓ Advanced prototype of Quantum QKD front-end modules